

Analytical Expression of Hydrodynamic Impact Pressure by Matched Asymptotic Expansion Technique

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Summary

Based upon critical review of Wagner's theory, an analytical solution of the two dimensional impact pressure has been derived by using matched asymptotic expansion technique. The solution is composed of three part: the linear part given from expanding flat plate solution, the nonlinear part valid in the splash region near the crossing point between the waterline and the body and the common part between both parts. The solution is compared to Wagner's results and is shown to give almost the same peak pressure but with different behaviors.

1. Introduction

Wagner's theory¹⁾ is widely known as the theory to give good prediction of peak pressure on a two-dimensional body in impact motion to the free surface. It may be said that various practical estimations²⁾ for the impact pressure free from air entrapping are based upon the theory to some extent. But there are some ambiguities in the formulation of mechanism of the peak impact pressure from hydrodynamic point of view.

To elaborate points, let us follow his arguments by using a wedge shaped body of constant deadrise angle plunging into the water at a constant velocity V for example. Basic assumptions adopted by Wagner are summerised into two, i. e.

- (1) that effects of gravity to the free surface can be neglected since perturbed fluid acceleration is much larger than that of gravity (gravity free) and
- (2) that the ratio of draft to breadth is so small that flow variation in the vertical direction along the body surface is negligibly smaller than of horizontal direction. (shallow draft)

Take a coordinate system fixed to the space as is shown in Fig. 1. It is easy to see that the free surface condition can be approximated by

$$\phi = 0 \quad \text{on} \quad y = 0 \quad (1)$$

according to the first assumption and the body condition can be moved to the discontinuity of the velocity potential on the free surface due to the second assumption as

$$\frac{\partial \phi}{\partial y} = -V, \quad |x| \leq a \quad \text{on} \quad y = 0. \quad (2)$$

Then the flow field is approximated in good

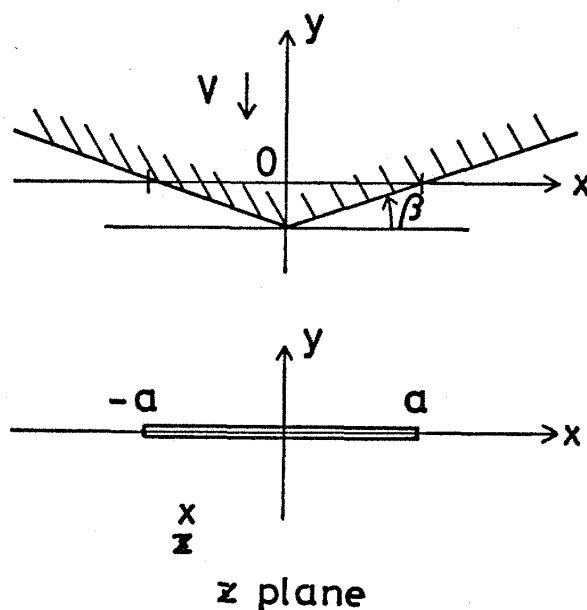


Fig. 1 coordinate system and body plan of wedge problem

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accuracy by that of flat plate in impulsive motion. It is easy to solve the flat plate problem and its velocity potential can be found elsewhere. The result is, on the plate ($y=0$),

$$\Phi = -V\sqrt{a^2 - x^2}. \quad (3)$$

In order to calculate pressure on the wedge from the potential (3), it is necessary to take into account of the fact that the breadth (a) is a function of time. Differentiating Φ with respect to t , we have

$$\frac{p}{\rho} = \frac{\partial \Phi}{\partial t} - \left(\frac{\partial \Phi}{\partial x} \right)^2 = \frac{Vaa_t}{\sqrt{a^2 - x^2}} - \frac{V^2a^2}{(a^2 - x^2)} \quad \text{on } y=0. \quad (4)$$

In computing expansion rate of the breath, effects of the surface elevation are taken into account and a ratio r is introduced as

$$r = \frac{a_t}{V \cot \beta} \quad (5)$$

where $r = \frac{\pi}{2}$ for the wedge.

Thus

$$\frac{p}{\rho} = V^2 r \cot \beta \frac{a}{\sqrt{a^2 - x^2}} - \frac{V^2a^2}{(a^2 - x^2)}. \quad (6)$$

The equation (5) presents following unresolved questions.

- (1) There are singularities of order of $1/x$ in the pressure as approaching the edges ($x=a, -a$). Therefore integration of the pressure for forces along the body is impossible.
- (2) The peak pressure occurs at the edge region where the linearity assumptions are not valid.
- (3) It is not appropriate to introduce quadratic terms in calculating pressure within a framework of the linear theory.

If we are to be consistent with linearity in the theory, the pressure has to be expressed by the first term in (4) alone, which is taken by some authors⁴⁾. But there will be no peak pressure once the quadratic term is dropped and the main feature of Wagner's theory will be lost.

Let us examine the origin of the peak pressure. According to the above formulation, the peak occurs as a result of balance between positive pressure due to momentum increase and negative pressure caused by infinite flow velocity around the corner. There is no such flow of infinite velocity at impact of the body to the water though. In reality, there appears formation of splash emanating from the edge regions of the bottom on the waterline.

Contemplating these facts, it will be said that questions on Wagner's solution came from that the linear formulation was applied to calculate impact pressure which occurs in the region where the linear assumptions are invalid. Clear explanation of splash flow is needed for more rational pressure formula. This problem can be seen as a variation of singular perturbation problem from a different view point. The splash region is a singular region where the ordinary linear assumption is violated. If we can formulate mechanism there and find a solution which continues smoothly asymptotically to the outer solution, we are to have uniformly valid solution.

2. General formulation

2.1 Statement of the problem

Let us take a flow field around a two-dimensional body plunging with velocity V into water. The body is assumed symmetric horizontally. The body surface may have any form under the

restriction that the x -component of the normal vector is of order ε , where ε is the tangent of the body at $y=0$ and is assumed very small.

Let $c_1, c_2, c_3 \dots$ be any non-negative constants of order $O(1)$. Then a body shape equation is defined as

$$y_0 = \varepsilon(|x_0| + b_1 x_0^2 + c_2 |x_0|^3 + c_3 x_0^4 + \dots) \equiv f(x_0) \quad (7)$$

Here the body shape and coordinates are shown in Fig. 2. (x_0, y_0) is a body fixed coordinate system. At the same time, we will take a space coordinate (x, y) of which origin is located at the center of the body waterline at time t (Fig. 2).

2.2 Expansion of boundary conditions

We will start our formulation with the assumptions made by Wagner. Namely, the gravity free condition for the free surface and the shallow draft assumption for the body boundary are taken as bases for the following discussion. Although zero potential on $y=0$ can be imposed in place of the free surface condition without further explanation as was shown in the last section, special care is needed in treating the body condition since there is a region where ordinary perturbation scheme breaks down.

The body shape equation at time t is expressed by using (x, y) as

$$y = \varepsilon f(x) - d(t) \quad (8)$$

where $d = d_0 + Vt$ is instantaneous draft.

The body condition in its exact form becomes

$$\varepsilon f_x \phi_x(x, y) - \phi_y(x, y) = d_t \quad (9)$$

where those with suffixes mean derivatives. Expanding around $y=0$ with respect to y , we have

$$\varepsilon f_x \{\phi_x(x, 0) + \varepsilon f \phi_{xy}(x, 0)\} - \{\phi_y(x, 0) + \varepsilon f \phi_{yy}(x, 0)\} = d_t \quad (10)$$

Equating the lowest order terms, the linearized boundary condition:

$$\phi_y(x, 0) = -d_t \quad \text{on } |x| \leq a(t), y = 0 \quad (11)$$

is obtained. It tells us that the problem is reduced to that of expanding flat plate in impulsive motion at a right angle. Although the flat plate solution describes well the impact flow in the almost all region, its x derivative ϕ_x becomes singular at $x = \pm a(t)$ like

$$\phi_x \sim \frac{2}{\sqrt{x^2 - a^2}} \quad (12)$$

As a result, the terms left behind in making linearised body condition (11) become order of unity in the edge regions $x \pm a = O(\varepsilon^2)$ and the body condition stop being valid there.

Let us examine the body condition in the edge regions. To make points clearer, the edge region near $x = -a$ is taken as an example. The same argument applies to the other region of

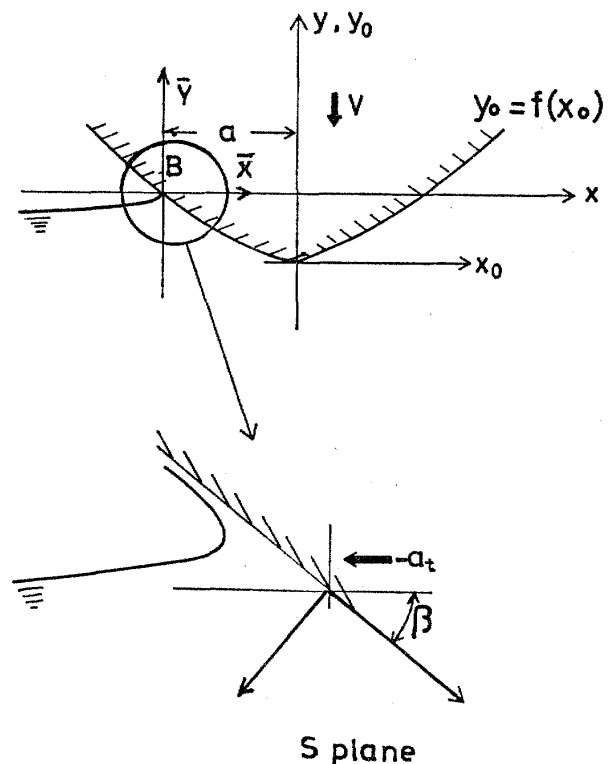


Fig. 2 view of 2-D body and its splash region

$x=a$ by slightly changing signs and symbols. Since the vertical velocity of the perturbed free surface by the body is small in the outer region, it may be assumed that the edge region moves away on the x axis at the velocity $-a_t(t)$. Introduce a new coordinate system $(\varepsilon^2 X, \varepsilon^2 Y)$ moving with the region as

$$\begin{aligned} x+a &= \varepsilon^2 \bar{X}, \\ y &= \varepsilon^2 \bar{Y}. \end{aligned} \quad (13)$$

Here $\bar{X}, \bar{Y}$ are stretched variables of $O(1)$.

Substituting them into the body equation (8), we have

$$\varepsilon^2 \bar{Y} = \varepsilon f(a + \varepsilon^2 \bar{X}) - d = \varepsilon f(a) + \varepsilon^3 f_x \bar{X} - d. \quad (14)$$

Since the first and the third terms cancel each other,

$$\bar{Y} = \varepsilon f_x(a) \bar{X} = -\tan \beta \cdot \bar{X}. \quad (15)$$

β is the deadrise angle at the instantaneous waterline.

Also from the exact body condition (9),

$$\tan \beta \cdot \phi_{\bar{X}}(\bar{X}, \bar{Y}) + \phi_{\bar{Y}} = 0 \quad \text{on} \quad \bar{Y} = -\tan \beta \cdot \bar{X}. \quad (16)$$

It is to be noted that the first order term is an order of unity in spite of $\tan \beta = O(\varepsilon)$ since ϕ_x is the order of $1/\varepsilon$. This condition shows us that the flow in the edge region can be approximated by the nonlinear flow of the inclined long flat plate in uniform stream with free surface. At the same time, another condition that there exist splash streaming away along the body surface will be required.

3. outer solution

As was shown in the last section, the flow field in the outer region is well described by the expanding flat plate in impulsive motion. It is easy to deduce its complex velocity potential w as

$$w(z) = iV(z - \sqrt{z^2 - a^2}). \quad (17)$$

The velocity potential is

$$\phi = \frac{1}{2} \{w(z) + \overline{w(\bar{z})}\} = -V \left\{ y + \frac{i}{2} (\sqrt{z^2 - a^2} - \sqrt{\bar{z}^2 - a^2}) \right\}, \quad (18)$$

where upper line denote conjugate complex.

The pressure distribution of the linear term becomes

$$\frac{p}{\rho} = -\frac{\partial \phi}{\partial t} = -V \frac{i}{2} \left\{ \frac{aa_t}{\sqrt{z^2 - a^2}} - \frac{aa_t}{\sqrt{\bar{z}^2 - a^2}} \right\} \quad (19)$$

Define non-dimensional pressure as

$$C_{p0} = \frac{p}{\frac{1}{2} \rho V^2} = -i r \cot \beta \left\{ \frac{a}{\sqrt{z^2 - a^2}} - \frac{a}{\sqrt{\bar{z}^2 - a^2}} \right\}. \quad (20)$$

Here r is the expansion rate of the waterline breadth defined by (5)

If we take into account of the effects of the water surface elevation due to the body motion, an integral equation introduced by Wagner is to be solved. The discontinuity of the velocity potential on $y=0$ induces a vertical velocity distribution on the free surface ($|x| > a, y=0$) as

$$v_y = \frac{V|x|}{\sqrt{x^2 - a^2}}. \quad (21)$$

Integrating v_y from time $t=0$ to t temporally, we have the elevation as

$$\eta = \int_0^t v_y dt = \int_0^x \frac{1}{r \cot \beta} \frac{x}{\sqrt{x^2 - a^2}} da. \quad (22)$$

Equating (7) with (22), an integral equation with respect to r is obtained. Assuming for $\frac{1}{r \cot \beta}$ a polynomial of the breadth, the solution is

$$\frac{1}{r \cot \beta} = \frac{2}{\pi} \varepsilon \left(1 + \frac{\pi}{2} c_1 a + 2 c_2 a^2 + \frac{3\pi}{4} c_3 a^3 \right). \quad (23)$$

The breadth (a) at t is given by solving a following equation:

$$Vt = \frac{2}{\pi} \varepsilon \left(a + \frac{\pi}{4} c_1 a^2 + \frac{2}{3} c_2 a^3 + \frac{3\pi}{16} c_3 a^4 \right). \quad (24)$$

At the same time, $\beta(t)$, instantaneous deadrise angle is given by

$$\tan \beta(t) = \frac{\partial y}{\partial x} = \varepsilon (a + 2 \beta_1 a + 3 \beta_2 a^2 + 4 \beta_3 a^3). \quad (25)$$

4. Inner solution (Splash region solution)

Let us examine the flow defined by the conditions (1), (16) and the splash condition that there exist splash running away along the body. The exact solution for it is not known and may be indeterminate because of the ill condition at infinity that upstream and downstream have different flow directions. However, except for its plate length, the conditions are the same for the planing plate problem of which solution is known. Thus if we make an asymptotic expansion of the planing plate solution in the splash region, it will satisfy above boundary conditions and will be taken as a inner solution.

4.1 Planning plate solution

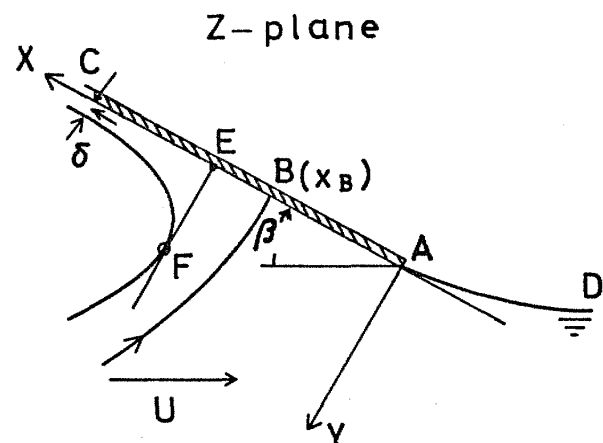
The flow field around a steady planing plate has been solved analytically and its velocity distribution will be found in literatures^{5,6,7)}. We will follow Kochin's procedure⁷⁾. Taking coordinate system as shown in Fig. 3 and uniform velocity U , velocity distribution in the fluid is given by

$$\frac{dw}{dZ} = U \frac{t-b}{1-bt + \sqrt{(1-b^2)(1-t^2)}}. \quad (26)$$

where $b = \cos \beta$,

And the relation between complex coordinate (Z) and parameter t is

$$\frac{\pi}{\delta} Z = \frac{1}{(1-b)} \left\{ (1+t)b - (1-b) \log \left(\frac{1-t}{2} \right) \right\}$$



t-plane

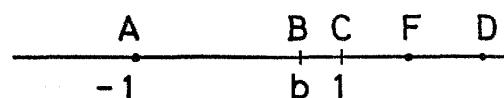


Fig. 3 general view of planing plate flow

$$+\sqrt{1-b^2} \left(\frac{\pi}{2} + \sin^{-1} t - \sqrt{1-t^2} \right) \} \quad (27)$$

where δ is the thickness of splash.

4.2 Inner solution

In making use of the expression (26) for the solution in the splash region, it is necessary to extract dominant terms near the stagnation point. For that purpose, eq. (27) is expanded about the splash root (x_B) assuming $\sin \beta$ is small. After some algebraic operations, we have

$$\frac{\pi}{\delta}(Z-x_B) = \frac{5+3b}{2(1-b)}(t-b). \quad (28)$$

Here δ may be taken unknown for our purpose that (26) is made to a candidate for the splash region solution. We will redefine alternative unknown constant A for further convenience as

$$A = \frac{5+3b}{2(1-b)} \frac{\delta}{\pi}, \quad (29)$$

which will be determined by matching with the outer solution. Rewriting (28) with (29), we have a new variable X' as

$$X' = \frac{x_B - Z}{A} = (b-t). \quad (30)$$

and the complex velocity expressed by the new variable is

$$\begin{aligned} u-iv &= U \frac{-X'}{1-b^2+bX'+\sqrt{(1-b^2)(1-b+X')(1+b-X')}} \\ &= U \frac{1-b^2+bX'-\sqrt{(1-b^2)(1-b+X')(1+b-X')}}{-X'} \end{aligned} \quad (31)$$

In order to have stretched picture near $X'=0$, take a new inner complex variable S as

$$S = \frac{X'}{(1-b^2)} \quad (32)$$

and substitute it into (31). We have an expansion of the velocity after dropping higher order terms

$$u-iv = U \frac{-S}{1+bS+k\sqrt{S+c}}. \quad (33)$$

where $k = \sqrt{1+b}$ and $c = 1/(1+b)$.

It is easy to see that (33) satisfies the body condition (16) since the imaginary part of complex velocity (33) becomes zero if we make S real.

In order to have an inner solution to be matched with the outer solution (20), the pressure distribution on the plate will be enough for our purpose. Define pressure coefficient as follows and substitute the velocity on the plate which is given by letting S real in (33).

Then we have

$$C_{pt} = \frac{p-p_\infty}{\frac{1}{2}\rho U^2} = 1 - \frac{1}{U}(u^2+v^2) = -\frac{\overline{SS} - (1+b\overline{S} - k\sqrt{\overline{S}+c})(1+b\overline{S} - k\sqrt{\overline{S}+c})}{SS} \text{ or}$$

$$= \frac{(1+bS+k\sqrt{S+c})(1+b\bar{S}+k\sqrt{\bar{S}+c})-S\bar{S}}{(1+bS+k\sqrt{S+c})(1+b\bar{S}+k\sqrt{\bar{S}+c})} \quad (34)$$

This is a candidate for the splash region solution with unknown constants U and A as well as unknown location of the origin of S . They will be determined by matching.

5. Matching and composite solution

Among unknown constants, U can be determined easily by inspection as

$$U = a_t = V_T \cot \beta. \quad (35)$$

and the origin of S can be taken at $x = -a$. In order to determine A , we will take following matching procedure:

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\cot \beta} (C_{pi} - C_{po}) \rightarrow 0. \quad (36)$$

instead of

$$\lim_{\varepsilon \rightarrow 0} (C_{pi} - C_{po}) \rightarrow 0. \quad (37)$$

We will define intermediate variable x_η in connection with outer and splash region variables as follows

$$\eta \bar{x}_\eta = a + z \quad (38)$$

$$\eta/\varepsilon^2 x_\eta = S, \quad (39)$$

$$\text{where } 0(\varepsilon^2) < 0(\eta) < 0(1).$$

Substituting these relations into (20) and the second line of (34), expanding with respect to ε and taking leading order terms of both solutions, we have

$$C_{po} \sim \frac{\sqrt{2}ar}{\sqrt{\eta}} \frac{1}{2} \left(\frac{1}{\sqrt{x_\eta}} + \frac{1}{\sqrt{\bar{x}_\eta}} \right) \cot \beta, \quad (40)$$

$$C_{pi} \sim b^2 \sqrt{1+b} r^2 \frac{\sqrt{A}}{\sqrt{\eta}} \left(\frac{1}{\sqrt{x_\eta}} + \frac{1}{\sqrt{\bar{x}_\eta}} \right) \cot \beta. \quad (41)$$

In order that both expansions are equal, A has to satisfy the relation:

$$A = \frac{a}{2b^4(1+b)r^2}. \quad (42)$$

This determines an unknown constant A only left behind. Now we can construct a composite expansion which is valid throughout the whole region by following procedure:

$$C_{pc} = C_{po} + C_{pi} - C_{pcom}, \quad (43)$$

where C_{pcom} is a common part in the both expansions at matching.

Applying this procedure to our case, the result valid for $x < 0$ is

$$C_{pc} = r \cot \beta \left[a \left\{ \frac{-i}{\sqrt{z^2 - a^2}} - \frac{1}{\sqrt{2a(z+a)}} + \frac{i}{\sqrt{\bar{z}^2 - a^2}} - \frac{1}{\sqrt{2a(\bar{z}+a)}} \right\} + r \cot \beta \frac{D}{S\bar{S}+D} \right]. \quad (44)$$

$$\text{where } D = k^2[|S+c| + \frac{1}{2}(S+\bar{S}+2c)] + k\{(1+bS)\sqrt{\bar{S}+c} + (1+b\bar{S})\sqrt{S+c}\}$$

$$S = \frac{(z+a)}{4(1-b^2)}.$$

It is seen that the maximum pressure occurs at $S=0$ ($x=-a$) and its magnitude is

$$C_p = r^2 \cot^2 \beta. \quad (45)$$

This value coincide with Wagner's result with negligible difference. The point where C_p becomes zero is located at

$$S = -c \quad (46)$$

$$\text{or} \quad x = -a - (1 - \cos \beta) a. \quad (47)$$

It is where local water elevation ends (E).

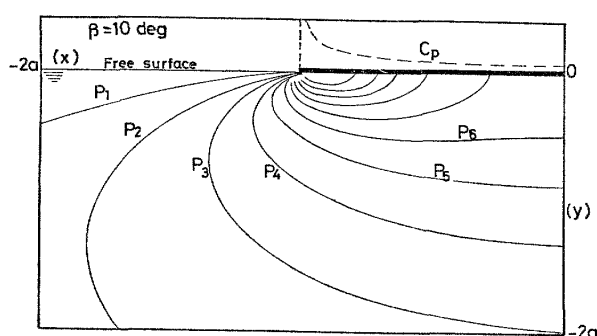


Fig. 4 equi-pressure contours due to the outer solution

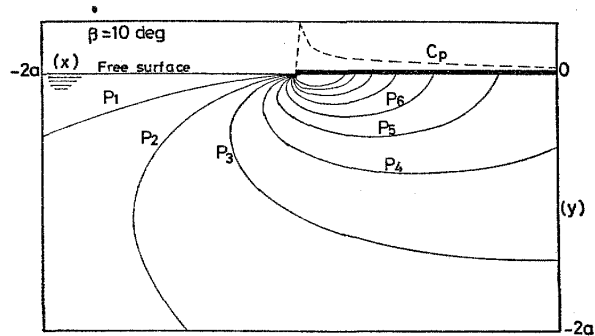


Fig. 5 equi-pressure contours due to the inner solution

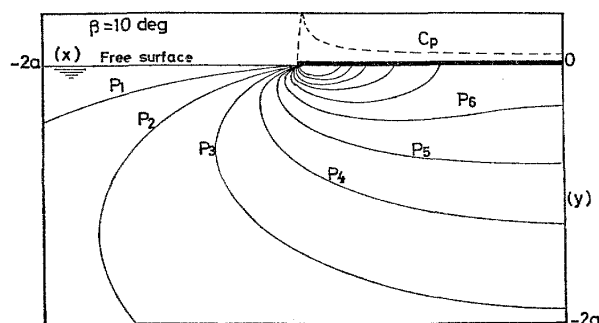


Fig. 6 equi-pressure contours due to the composite solution

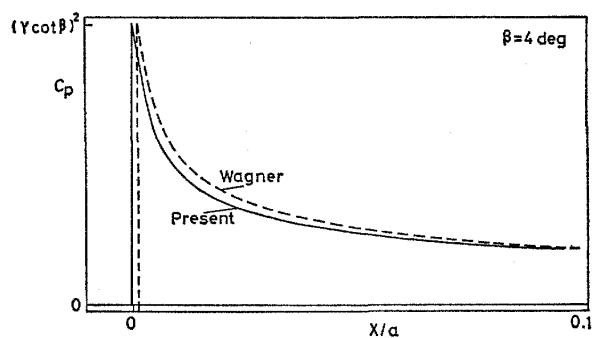


Fig. 7 spatial impact pressure distribution

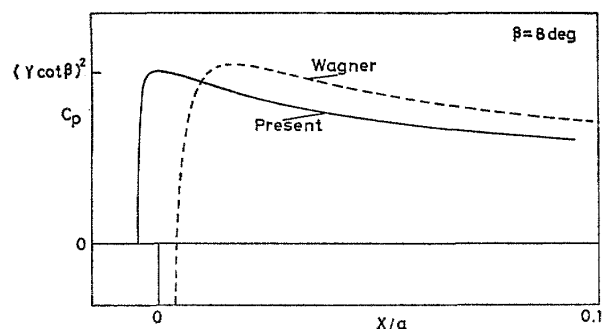


Fig. 8 spatial impact pressure distribution

gives the surface pressure over the body. It is seen from the behavior of the contours and surface pressure that there is a singularity at $x=-a$. Fig. 5 shows the pressure contours com-

6. Numerical examples

Let us examine contributions from each components comprising composite solution (44) for a wedge-shaped body of constant dearse angle. ($\beta=10$ deg.) Fig. 4 shows pressure contours given by the first and second terms in the braces of (44), which represent the contribution by the outer solution. Also shown is a dotted line which

puted from the second term in the square bracket of (44), which represents a component due to the inner solution. It is clear that the pressure has a peak near $x=-a$ and decreases as approaching to the center part. Fig. 6 shows pressure contours and surface pressure (dotted line) given by the composite solution (44). It is seen that the pressure has the similar behavior to the inner solution near $x=-a$ while it comes closer to the outer solution in the center part on the body surface. These behaviors show that the solution has been constructed correctly.

Computed results by the present method are compared to other data. Fig. 7 to 8 show the spatial impact pressure distributions over the surface of wedges of constant deadrise angle. Comparing to Wagner's results drawn by the dotted line, present calculations show no singular behavior at the edge but go to zero in natural manner. It is seen that both calculations give almost the same maximum pressure as already seen in the previous section. As deadrise angle increases, present method predicts that the increase of pressure occurs at the negative x , whereas Wagner's calculation tells it occurs at larger positive x . This difference comes from the fact that Wagner's method fails to take the effects of local surface elevation into account correctly.

7. Concluding remarks

Based upon the critical review of Wagner's theory on the hydrodynamic impact pressure, a new analytical formulation of the impact pressure has been derived by means of matched asymptotic expansion. It was shown that the present method succeeded in explaining the mechanism of the impact pressure on the plunging two dimensional body. Since this method seems to have capability to handle more general cases such as three dimensional impact, efforts along that line would give us interesting results.

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References

- 1) H. Wagner: Über Stoss- und Gleitvorgänge an der Oberfläche von Flüssigkeiten, band 4, heft 4, ZAMM (1932).
- 2) A. B. Stavovy and S-L Chuang: Analytical Determination of Slamming Pressures for High-Speed Vehicles in Waves, Vol 13, No. 4, J. of Ship Res. (1976).
- 3) J. D. Cole: Perturbation Methods in Applied Mathematics, Blaisdell (1968)
- 4) Yoichi Hirano: Hydrodynamic Impact, J. of the Japan Society for aeronautical and Space Sciences, No. 214, Vol. 19 (1971)
- 5) L. M. Mine-Thomson: Theoretical Hydrodynamics, Macmillan (1955)
- 6) L. I. Sedov: Two-dimensional Problems in Hydrodynamics and Aerodynamics, Interscience (1950)
- 7) N. E. Kochin, I. A. Kibel and N. V. Roze: Theoretical Hydromechanics, Interscience (1955)